Student User Manual & Registration Guide

Welcome to Electify! This system manages your elective course selections with zero waitlists, live seat counts, and instant tamper-evident receipt generation. Follow this guide to register seamlessly.

1. Access & Account Authentication

STEP 1 Signing in to the Student Portal

1. Open your web browser (Chrome, Edge, Safari, Firefox) and navigate to: `http://localhost:3000/login` (or your college Electify URL).
2. Enter your **SRM Register Number** (e.g., `RA2411003010001`) or official SRM email.
3. Enter your password and click **Sign in to Electify**.

Security Policy: Single-Session Sentinel
You are allowed only **one active session at a time**. If you log in on another device or a separate browser window, your previous session is immediately revoked. Never share your credentials with classmates.

2. Exploring Courses Before the Window Opens

STEP 2 Subject Explorer & Preview Mode

Before the registration countdown expires, you can browse all offered electives:

1. Click **Subjects** in the navigation sidebar (or visit `/dashboard/subjects`).
2. Browse course descriptions, syllabi, course credits, and prerequisite criteria.
3. Inspect the total capacity allocated for each subject.

Pro-Tip: The Subject Explorer is read-only and does not reserve seats. Prepare your top choices in advance so you can submit promptly once the window unlocks.

3. Live Registration & Atomic Seat Locking

STEP 3 Synchronized Countdown Clock

When an elective registration event is scheduled, navigate to `/countdown`. The countdown timer is **synchronized directly with the university time server (NTP-drift corrected)**, ensuring absolute fairness across all student devices.

STEP 4 Selecting Your Electives

1. When the clock hits zero, click **Enter Registration** (or visit `/dashboard/electives`).
2. View the available courses under each **Elective Basket**.
3. Observe the live seat indicators:
   - **GREEN:** Seats comfortably available (e.g., 45 / 60 remaining).
   - **AMBER:** Filling quickly (less than 10% remaining).
   - **RED:** Fully booked (0 seats remaining).
4. Select your preferred course in each required category and click **Submit Registration**.

How Atomic Row-Locking Works:
Electify uses PostgreSQL database row-level locking (`SELECT ... FOR UPDATE`). When you submit, the server checks seat availability and decrements the counter in a single atomic transaction. No overbooking is mathematically possible.

4. Confirmation & Official PDF Receipt

STEP 5 Downloading Your Official Receipt

1. Upon submission, you are immediately routed to `/dashboard/success`.
2. Your unique **Official Receipt ID** will be displayed in the format: `ELC—<YEAR>-<ODD|EVEN>-<SEQUENCE_NUMBER>` (e.g. `ELC-2026-EVEN-00421`).
3. Click **Download PDF Receipt** to save your official verification document. Retain this PDF for your academic records and tutor verification.

5. Troubleshooting & Frequently Asked Questions

Issue / Question	Solution / Protocol
Forgot Password / Locked Out	Students cannot self-reset passwords to prevent account takeover. Contact your **Class Tutor** directly; they will verify your identity and authorize a 15-minute PIN reset.
Course filled while selecting	If another student claimed the last seat a millisecond before you, the system notifies you instantly. Choose an alternative elective in that basket and submit immediately.
Can I modify my choice?	Modifications are allowed only if the Course Coordinator enabled *"Allow Modifications"* and the registration window has not reached its deadline.